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End Semester Examination 2025

Name of the Course: B. Tech (EE)
Name of the Paper – Power Electronics
Paper Code: TEE 601

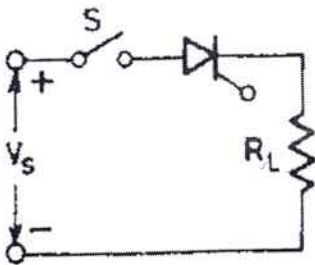
Semester: VI

Time: 3Hours

Maximum Marks: 100

Note: -

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q. 1	(20 Marks)	
(a)	Describe the various anode voltage ratings as applicable to an SCR. Indicate these voltage ratings on a relevant voltage waveform.	CO1
	OR	
(b)	Define latching and holding currents as applicable to an SCR. Show these currents on its static V-I characteristics. Describe, with a sketch, the effect of gate current on the forward breakover voltage of an SCR. Also discuss the methods of turning on of a thyristor with its gate disconnected.	CO1
	OR	
(c)	The Fig. shows a thyristor controlling the power in a load resistance R. The supply voltage is 240 V dc and the specified limits for di/dt and dv/dt for the SCR are 50 A/ μ sec and 300 V/ μ sec respectively. Determine the values of the di/dt inductance and the snubber circuit parameters R, and C.	CO1
		
Q. 2	(20 Marks)	
(a)	For a single-phase one-pulse controlled converter system, sketch waveforms for load voltage and load current for (i) RL load and (i) RL load with freewheeling diode across RL. From a comparison of these waveforms, discuss the advantages of using a freewheeling diode.	CO2
	OR	
(b)	A single-phase 230 V, 1kW heater is connected across 1-phase, 230 V, 50 Hz supply through an SCR. For firing angle delays of 45° and 90°, calculate the power absorbed in the heater element.	CO2
	OR	

(c)	Describe the operation of a single-phase two-pulse mid-point converter with relevant voltage and current waveforms. Discuss how each SCR is subjected to a reverse voltage equal to double the supply voltage in case turns ratio from primary to each secondary is unity.	CO2
Q. 3	(20 Marks)	
(a)	A single-phase full bridge inverter has RLC load of $R = 4 \text{ ohm}$, $L = 53 \text{ mH}$ and $C = 155 \text{ } \mu\text{F}$. The dc input voltage is 230 V and the output frequency is 50 Hz. (a) Find an expression for load current up to fifth harmonic. Also, calculate (b) rms value of fundamental load current. (c) the power absorbed by load and the fundamental power.	CO3
	OR	
(b)	A single-phase full-bridge inverter may be connected to a load consisting of (a) R (b) RL. For all these loads, draw the load voltage and load current waveforms under steady operating condition. Discuss the nature of these waveforms.	CO3
	OR	
(c)	Discuss the principle of working of a three-phase bridge inverter with an appropriate circuit diagram. Draw phase and line voltage waveforms on the assumption that each thyristor conducts for 120° and the resistive load is star-connected. The sequence of firing of various SCRs should also be indicated in the diagram.	CO3
Q. 4	(20 Marks)	
(a)	For the basic dc to dc converter, express the following variables as functions of V_s , R and duty cycle, in case load is resistive. (a) Average output voltage and current (b) RMS value of the output voltage	CO4
	OR	
(b)	Explain, with appropriate waveforms, the different control strategies used for obtaining variable output voltage from a dc chopper. Which one of these is preferred over the other and why.	CO4
	OR	
(c)	A step-up chopper has input voltage of 220 V and output voltage of 660 V. If the non-conducting time of thyristor-chopper is $100 \text{ } \mu\text{s}$, compute the pulse width of output voltage. In case pulse width is halved for constant frequency operation, find the new output voltage.	CO4
Q. 5	(20 Marks)	
(a)	What is a cycloconverter? Enumerate some of its industrial applications. Describe the operating principle of single-phase to single-phase step-up cycloconverter with the help of mid-point type configurations. Illustrate your answer with appropriate circuit and waveforms. The conduction of various thyristors must also be indicated on the waveforms.	CO5
	OR	
(b)	For a single-phase voltage controller feeding a resistive load, draw the waveforms of source voltage, gating signals, output voltage, source and output currents and voltage across one SCR. Describe its working with reference to the waveforms drawn.	CO5
	OR	
(c)	A single-phase voltage controller has input voltage of 230 V, 50 Hz and a load of $R = 15 \text{ ohm}$. For 6 cycles on and 4 cycles off, determine (a) rms output voltage, (b) input pf.	CO5